

# New Attention Ensemble-Based Fault Diagnosis for Industrial Processes Under Imbalanced Finite Data

Yaoqian Zhu<sup>✉</sup> and Ridong Zhang<sup>✉</sup>

**Abstract**—During industrial processes operation, fault diagnosis needs to rely on finite and imbalanced samples for learning because the equipment is usually in a normal operation state and the probability of faults is low. For this issue, we propose a new attention ensemble algorithm, DMSAM-OAdB. First, the multiscale sparse attention module (MSAM) extracts and fuses feature information from different scales to establish the correlation among features and adaptively adjusts the feature weights by soft thresholding operation to emphasize the critical features while filtering the redundant information. Subsequently, a deep network model (DMSAM) constructed by stacking multiple convolutional layers and MSAMs is iteratively trained as the base classifier for the optimized AdaBoost algorithm (OAdB). The OAdB dynamically increases the weights of misclassified samples while mitigating class imbalance by randomly zeroing out the weights of a certain number of correctly classified majority class samples. Finally, the improved focal loss (IFL) function enhances the model's discrimination capability for minority class samples. Experiments on the industrial coking furnace validate the reliability and superiority of the proposed methodology.

**Index Terms**—AdaBoost, attention mechanism, fault diagnosis, loss function, multiscale network.

## I. INTRODUCTION

**I**N INDUSTRIAL processes, factors such as aging and wear of parts, leakage and clogging of pumps, and changes in ambient temperature and humidity inevitably affect the system operation and lead to malfunctions. Therefore, leveraging intelligent fault diagnosis to accurately determine the type of faults is of great significance in enhancing the reliability and ensuring production safety [1].

Traditional fault diagnosis methods based on physical models, statistical process control, and process signal analysis usually rely on assumptions and domain expertise, which are insufficient to cope with complex challenges posed by modern industry. Consequently, data-driven methods based on deep learning, such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and deep auto encoders (DAEs), have gained significant attention by virtue of their ability to extract fault features, adapt to diversified fault modes, and reduce reliance on expert knowledge. For instance, Ye and Ju [2] proposed a multilevel feature fusion network using multiscale convolution and dynamic feature selection. This

network extracts and integrates multiscale and multilevel fault features using convolution kernels of different sizes, which enables intelligent fault diagnosis for bearings. Liu et al. [3] proposed a model fusion framework using multiscale residual CNN and LSTM, which effectively extracts multiscale features and temporal information from industrial process data and achieves higher accuracy. Fault diagnosis is essentially a pattern classification issue, so the first task is to extract rich key information from the input signal. In recent years, the attention mechanism has also gained popularity due to its advantages of extracting global information and rationally allocating resources. Some notable results have been achieved by introducing it to enhance fault classification ability. Wan et al. [4] developed a novel multisensor feature-level information coupling model for bearing fault diagnosis using the mutual attention mechanism, which allows each layer of the feature extraction module to comprehensively analyze the coupling relationship between representative features in multisensor data. Wei et al. [5] proposed a target Transformer model that employs a target attention mechanism to effectively capture the long-term dependency characteristics of chemical processes and significantly improves the accuracy.

While deep learning has significantly promoted intelligent fault diagnosis, it also brings new challenges. Deep neural network models rely on a large amount of historical data and often assume that the data distribution is balanced. However, industrial processes are mostly in the normal operating state with faults occurring infrequently. As a result, the number of fault samples collected is extremely finite and significantly smaller than the normal samples. Additionally, attention-based models, particularly pure attention architectures such as Transformers, often require large-scale data to improve the model's classification and generalization capabilities, as they are different from models like CNN that inherently incorporate inductive biases such as local invariance and translational equivariance. Inductive bias can be understood as the introduction of certain rules or assumptions to impose certain constraints on model selection, which is similar to a kind of priori information [6]. When faced with finite or imbalanced datasets, insufficient inductive bias may impair the model's ability to learn effective patterns, potentially leading to classification bias, the tendency to favor majority classes and the neglect of minority class features, thereby reducing the classification accuracy for minority faults. To address these issues, we integrate CNN with attention mechanisms to introduce inductive bias, thereby reducing the impact of finite and imbalanced data.

Received 19 February 2025; revised 21 April 2025; accepted 9 May 2025. Date of current version 16 January 2026. The Associate Editor coordinating the review process was Dr. Pedro Oliveira Conceca. (Corresponding author: Ridong Zhang.)

The authors are with the Information and Control Institute, Hangzhou Dianzi University, Hangzhou 310018, China (e-mail: gz9513@163.com; zhangridong@hdu.edu.cn).

Digital Object Identifier 10.1109/TIM.2025.3575980

1557-9662 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

Within complex industrial environments, process data are often accompanied by redundant information. For this reason, Jiang et al. [7] proposed the digital twin auxiliary adaptive sparse attention network using soft thresholding to dynamically filter redundant sequences. Zhang et al. [8] designed a 2-D sparse low-rank model to extract the sparsity and periodic self-similarity within groups in bearing fault signals. However, these methods generally rely on sufficient data to train the network and fail to consider the scarcity of fault samples in real industrial processes. Moreover, most of them adopt single-scale CNNs for feature extraction, overlooking the variations in fault pattern characteristics across different spatial and temporal scales.

In the field of industrial fault diagnosis, methods for dealing with data imbalance can be broadly categorized into two main types: one is the adjustment of the data itself, such as oversampling to augment the number of minority class samples [9] and undersampling to remove majority class samples [10]; the other is the design at the algorithm level, such as cost-sensitive learning [11] and ensemble learning [12]. Duan et al. [13] proposed an improved SMOTE algorithm that addresses class imbalance in mechanical fault diagnosis by generating new minority class samples using the combination of geometric centers and average radii of existing samples. Zhou et al. [14] employed a hybrid sampling strategy that integrates random sampling with SMOTE to balance the number of samples across different fault categories. However, these data-level methods directly operate on raw data, which may cause information loss or introduce noise. In contrast, the algorithmic-level approach can focus on minority class samples more effectively by introducing an adjustment mechanism to balance the class weights while avoiding overalteration of the majority class samples and is therefore more relevant to real-world application needs.

To address the above problems, this article proposes a new attention ensemble algorithm for fault diagnosis under conditions of limited and unbalanced samples in industrial processes. The main contributions are as follows.

- 1) A new multiscale sparse attention module (MSAM) that establishes and integrates long-range dependencies and local feature relationships for feature mapping at different scales is proposed. Also, an adaptive soft-thresholding operation is introduced to filter the attention weight sequence while removing the redundant information interference and focusing more on critical features.
- 2) An improved focus function is designed to enhance the model's ability to distinguish between minority class samples and fuzzy area samples.
- 3) The optimized AdaBoost algorithm (OAdB) is introduced, which not only focuses more on misclassified samples but also reduces the influence of redundant information in the majority class by randomly setting a certain proportion of correctly classified majority class sample weights to zero during each iteration.

## II. PROPOSED METHOD

Considering the difficulty in obtaining a large number of fault samples in actual industrial operation, this

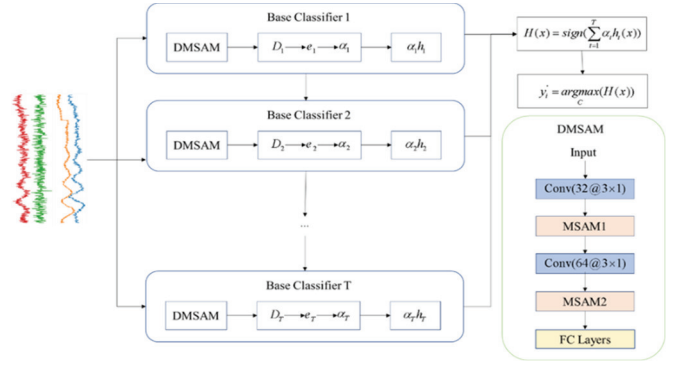


Fig. 1. DMSAM-OAdB model framework diagram.

article proposes an improved attention ensemble algorithm, DMSAM-OAdB, to accurately identify and classify finite and imbalanced fault samples. The overall framework of DMSAM-OAdB is shown in Fig. 1.

### A. Multiscale Sparse Attention Module

The operating status of the systems can be affected by many factors, including internal temperature, pressure, flow rate and external ambient interference. These factors can result in distinct fault patterns across different spatial and temporal scales [15]. This also implies that single-scale convolution operations are insufficient to fully extract all fault-related information. Generally, larger scales can generate a smoother representation for capturing low-frequency information, while small scales offer a higher detail resolution ability and highlight changes in high-frequency features [16]. Therefore, we adopt a multiscale mechanism by constructing convolution layers of different sizes in parallel to comprehensively capture and analyze fault features. Initially, the input feature map  $X_{in} \in \mathbb{R}^{h \times w \times ch}$  passes through a  $1 \times 1$  convolutional layer to achieve information integration and feature dimensionality reduction, where  $h$ ,  $w$ , and  $ch$  represent the length, width, and number of channels of the input feature map, respectively. Then,  $m$  convolution kernel groups  $K_1, K_2, \dots, K_m$  of different sizes are used to extract multiscale features from the output feature map after dimensionality reduction. Each convolution kernel group  $K_m$  has  $n$  convolution kernels, i.e.,  $K_m = [K_m^1, K_m^2, \dots, K_m^n]$ ,  $K_m^n \in \mathbb{R}^{s_m \times 1}$ , where  $s_m$  represents the convolution kernel size. The specific process is expressed as follows:

$$\tilde{U} = K^{1 \times 1} * X_{in} \quad (1)$$

$$U_m^n = K_m^n * \tilde{U} \quad (2)$$

$$U_m = [U_m^1, U_m^2, \dots, U_m^n] \quad (3)$$

$$\hat{U} = \sum_{i=1}^m U_i \quad (4)$$

where  $\tilde{U}$  represents the output of the  $1 \times 1$  convolution layer,  $K^{1 \times 1}$  represents the convolution kernel of the  $1 \times 1$  convolution layer,  $*$  represents the convolution operation,  $U_m$  represents the convolution output of the convolution kernel group of the corresponding size, and  $\hat{U} \in \mathbb{R}^{h' \times w' \times n}$  represents the fused feature obtained by adding the convolution outputs of each scale.

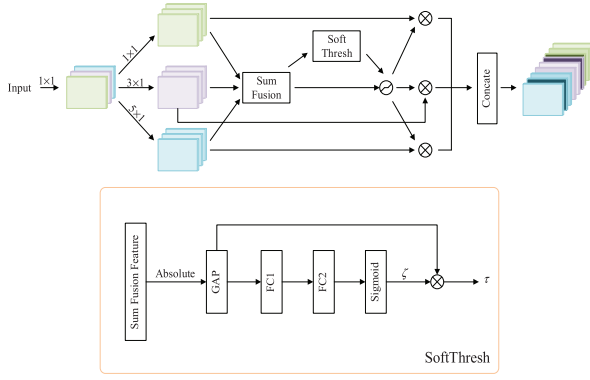


Fig. 2. MSAM structure.

Given the varying importance of information at different scales and the impact of redundant information on model robustness, we adopt a soft thresholding dynamic selection attention mechanism so that the model can adaptively adjust the contribution of feature maps at different scales after enhancing the precision of feature selection. The specific definition of the soft threshold function is shown in (5), where  $x$  is the input feature,  $\tau$  represents the threshold, and  $\text{SoftThresh}(\cdot)$  represents the output map after soft thresholding. By setting different  $\tau$  values, the soft thresholding operation can compress features within a specific range to zero values [17]. This article adaptively adjusts the threshold through the deep model architecture

$$\text{SoftThresh}(x; \tau) = \begin{cases} x - \tau, & x > \tau \\ 0, & -\tau \leq x \leq \tau \\ x + \tau, & x < -\tau. \end{cases} \quad (5)$$

The complete MSAM structure is shown in Fig. 2. As shown in the bottom part of Fig. 2, the fused multiscale features are compressed into a 1-D vector after absolute value operation and global average pooling (GAP) layer processing and passed to two fully connected (FC) layers.

The output of the FC layers is scaled to the range of (0, 1) as follows:

$$\zeta_l = \frac{1}{1 + e^{-z_l}} \quad (6)$$

where  $z_l$  represents the output map of the  $l$ th neuron and  $\zeta_l$  denotes the obtained scaling parameter for the  $l$ th feature. The threshold  $\tau$  can be obtained by multiplying the scaling parameter with the output of the GAP layer

$$\tau_l = \zeta_l \cdot \text{average}_{h', w'} |x_{h', w', l}| \quad (7)$$

where  $\tau_l$  refers to the calculated threshold of the  $l$ th channel and  $h'$ ,  $w'$ , and  $l$  represent the length, width, and channel index of the input feature map of the GAP layer, respectively. Subsequently, by adaptively extracting information of different spatial scales, the attention weight matrix  $A$  is dynamically generated, which is calculated as follows:

$$A = [A_1, A_2, \dots, A_m] \\ = \sigma(\text{SoftThresh}(\hat{U}; \tau))$$

$$= \frac{1}{1 + e^{-\text{SoftThresh}(\hat{U}; \tau)}}. \quad (8)$$

Finally, the weights corresponding to the feature maps of each scale are fused with the original feature maps to obtain the output features  $V$  of the MSAM, where  $\otimes$  represents the element-by-element multiplication

$$V = [V_1, V_2, \dots, V_m] \quad (9)$$

$$V_m = U_m \otimes A_m. \quad (10)$$

### B. Optimized AdaBoost Algorithm

The ensemble learning can be divided into two categories: parallelization methods and serialization methods. In parallelization methods, the base classifiers are independent of others and have no significant dependencies. Therefore, they can be generated and trained in parallel. Comparatively, the base classifiers in the serialization methods have strong dependencies and the subsequent classifiers are usually generated by adjusting the results of the previous classifier. When a base classifier misjudges a data sample, the remaining base classifiers can correct it. This mechanism not only improves the performance of the overall model but also effectively reduces the possibility of overfitting, thereby enhancing the reliability and stability of the model [18], [19].

AdaBoost is a classic implementation of the typical serialization algorithm boosting with the advantage of adaptively adjusting sample weights. For updating sample weights, it increases the weights of misclassified samples and reduces the weights of correctly classified samples. Through this step-by-step optimization, the model can more accurately identify challenging samples. When assigning base classifier weights, the weights assigned to each classifier are inversely proportional to its classification error rate. The classifier with a lower error rate contributes more to the final decision. Finally, through the weighted fusion method, AdaBoost integrates multiple base classifiers into a strong classifier. Inspired by the BalanceCascade algorithm [20] that gradually reduces the majority class samples predicted correctly to balance the training set, we add this zeroing mechanism to AdaBoost to optimize it. In each iteration, the OADB algorithm will zero the weights of the majority class samples that are correctly classified and do not exceed the minimum number of class samples. This further alleviates the class imbalance while maintaining the integrity of the data distribution and reduces the excessive dependence of the strong classification model on the correctly classified samples.

This article employs a deep network model based on the MSAM as the base classifier for the OADB. The specific implementation is as follows.

1) Construct a dataset  $D_{tr} = \{(x_i, y_i)\}_{i=1}^N$ , where  $x_i$  represents the input data,  $y_i$  represents the corresponding label, and  $N$  represents the number of samples. At the initial stage, each sample is assigned an equal weight to be considered of the same importance

$$D_i = \frac{1}{N}. \quad (11)$$

2) During the iterative training of  $t = 1, 2, \dots, T$ , the base classifier  $h_t$  is trained using the current sample weight

distribution to minimize the classification error as much as possible. The weighted classification error  $e_t$  can be expressed as follows:

$$e_t = \sum_{i=1}^N D_t(i) I(h_t(x_i) \neq y_i) \quad (12)$$

where  $I()$  represents the indicator function. If the classification is wrong,  $I(h_t(x_i) \neq y_i) = 1$ ; otherwise, it is equal to zero. Since industrial fault diagnosis is a multiclassification issue, the SAMME.R algorithm is employed to adjust the base classifier weight  $\alpha_t$

$$\alpha_t = \frac{1}{2} \ln \frac{1 - e_t}{e_t} + \ln(R - 1) \quad (13)$$

where  $R$  represents the number of categories. The smaller the value of  $e_t$ , the better the performance of base classifier  $h_t$ , resulting in a larger assigned weight and a greater contribution to the final prediction result.

Then, the training sample weight  $D_{t+1}$  is updated according to the performance of each base classifier and normalized

$$D_{t+1}(i) = D_t(i) \cdot e^{(-\alpha_t y_i h_t(x_i))} \quad (14)$$

$$D_{t+1}(i) = \frac{D_{t+1}(i)}{\sum_{j=1}^N D_{t+1}(j)}. \quad (15)$$

If sample  $x_i$  is correctly classified ( $y_i = h_t(x_i)$ ), (14) will reduce its weight; otherwise ( $y_i \neq h_t(x_i)$ ), the weight of the sample will be increased. During the learning process for class-imbalanced tasks, minority class samples are more likely to be misclassified, leading to a continuous increase in their weights through iterations. This mechanism enhances the model's ability to recognize minority classes and mitigates the performance degradation caused by imbalanced sample distributions.

After obtaining the updated sample weights, a random zeroing mechanism is used to set the weights of a certain proportion of correctly classified majority class samples  $M_t$  to zero,  $M_t = \{i | h_t(x_i) = y_i = y_{\text{majority}}, i \leq \text{Count}(\text{minority})\}$ . An excessively high proportion of zeroing weighted samples may lead to insufficient training data, thereby degrading the learning performance and generalization ability of the base learners. Therefore, in each new training round, the weights of correctly classified majority class samples are set to zero at a ratio of 0.1. Moreover, the number of samples with zero weight is constrained not to exceed the number of minority class samples

$$D_{t+1}(i) = \begin{cases} 0, & i \in M_t \\ D_{t+1}(i), & \text{else.} \end{cases} \quad (16)$$

Upon completing the predetermined number of iterations, the final output  $H(x)$  of the DMSAM-OAdB can be expressed as

$$H(x) = \text{sign} \left( \sum_{t=1}^T \alpha_t h_t(x) \right). \quad (17)$$

Specifically, the final classification result is obtained by aggregating the predictions of all base classifiers through a weighted summation using their assigned weights  $\alpha_t$ . This ensemble approach effectively reduces both bias and variance

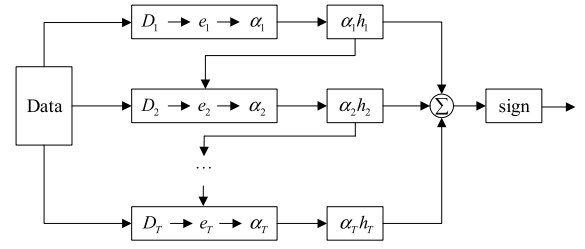


Fig. 3. OAdB model structure.

and gives a more robust and reliable model. Fig. 3 shows the specific structure of the OAdB model.

### C. Model Classification and Optimization

In the classification stage, the FC layer is responsible for integrating the obtained high-dimensional features and mapping them to the category space. Then, the abstract representation is converted into the probability distribution of each category by combining the softmax activation function

$$P_i = \text{softmax}(F_i) = \frac{e^{F_i}}{\sum_{k=1}^R e^{F_k}} \quad (18)$$

where  $F_i$  represents the output of the  $i$ th neuron in this layer and  $P_i$  is the probability of  $F_i$ . The predicted fault label  $y'_i$  is determined by selecting the category associated with the maximum probability

$$y'_i = \underset{C}{\text{argmax}}(P_i). \quad (19)$$

To quantify the difference between model predictions and true targets, a loss function is required to guide the gradual adjustment of model parameters. Specifically, deep learning employs the backpropagation algorithm to compute the gradients of the loss function and iteratively updates the weight values. This aims to progressively reduce prediction errors and improve model performance during continuous training. In multiclass classification, the cross-entropy (CE) loss function is typically used to evaluate the matching degree between probability distributions. However, when dealing with imbalanced data samples, the CE function overlooks class distribution differences, resulting in a bias toward majority class samples and difficulty in recognizing difficult-to-classify samples. Therefore, the focal loss function [21]  $L_{FL}$  is proposed to solve this problem

$$L_{FL} = \frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C \begin{cases} -\beta(1 - P_{i,c})^\gamma \log(P_{i,c}), & y_{\text{true},i} = 1 \\ -(1 - \beta)P_{i,c}^\gamma \log(1 - P_{i,c}), & y_{\text{true},i} = 0. \end{cases} \quad (20)$$

To further enhance the model's sensitivity to the intermediate prediction regions, the improved focal loss (IFL) function is adopted as the loss function for the DMSAM-OAdB model as follows:

$$L_{IFL} = \frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C \begin{cases} -\beta \tanh(1 - P_{i,c})^\gamma \log(P_{i,c}), & y_{\text{true},i} = 1 \\ -(1 - \beta) \tanh(P_{i,c})^\gamma \log(1 - P_{i,c}), & y_{\text{true},i} = 0. \end{cases} \quad (21)$$



TABLE I  
COKING FURNACE PROCESS FAULT DESCRIPTION

| NO. | Fault Description                     | Type  |
|-----|---------------------------------------|-------|
| 1   | Step signal, oxygen content 2         | Step  |
| 2   | Step signal, oxygen content 1         | Step  |
| 3   | Slow drift signal, temperature B      | Drift |
| 4   | Random signal, temperature B          | Other |
| 5   | Step signal, thermal efficiency point | Step  |

The focal loss function weakens the dominant effect of normal samples through the combined weight adjustment mechanism of the modulation factor  $(1 - p_i)^\gamma$  and the category weight  $\beta$ , while the tanh function further enhances this effect. The tanh function compresses the prediction weights of easily classified samples close to extreme values and amplifies the prediction weights of difficult-to-classify samples in intermediate ambiguous regions, thereby reducing the interference of easily classified samples on the overall loss and strengthening the model's attention to boundary samples in the fuzzy area. Additionally, the tanh function has the effect of smoothing extreme weight changes. This smooth nonlinear characteristic helps to alleviate the gradient oscillation caused by excessive or unstable gradient fluctuations. Consequently, it contributes to better convergence and robustness.

#### D. Fault Diagnosis Steps

*Step 1 (Preprocessing Data):* Mark, randomly shuffle, and normalize the acquired data. Then, the standardized dataset is divided into training and testing sets.

*Step 2 (Model Training):* Input the training set data into the DMSAM-OAdB model. The MSAM module extracts and fuses the features with different sizes of receptive fields, refines key information, and removes redundant information interference through soft thresholding and sigmoid activation. The OAdB dynamically adjusts the weights of samples and classifiers in each new iteration to minimize the impact of limited sample and class imbalance while improving the model's overall ability to recognize difficult-to-classify samples.

*Step 3 (Model Optimization):* The Adam optimizer is utilized to update the model parameters along the gradient direction to minimize the IFL loss function, thereby obtaining and saving a strong classification model.

*Step 4 (Model Verification):* Input the testing set data into the saved strong classification model for fault diagnosis.

### III. EXPERIMENTAL RESEARCH

#### A. Data Introduction

Industrial coking furnace F101/3 is a heating furnace, as shown in Fig. 4. It is mainly designed to rapidly heat the temperature of raw material for deep processing of crude oil. Its fuel source is primarily self-produced high-pressure gas. The operation of the coking furnace includes three key stages: high-temperature heating of feedstock, separation of components, and cracking to generate target products [22].

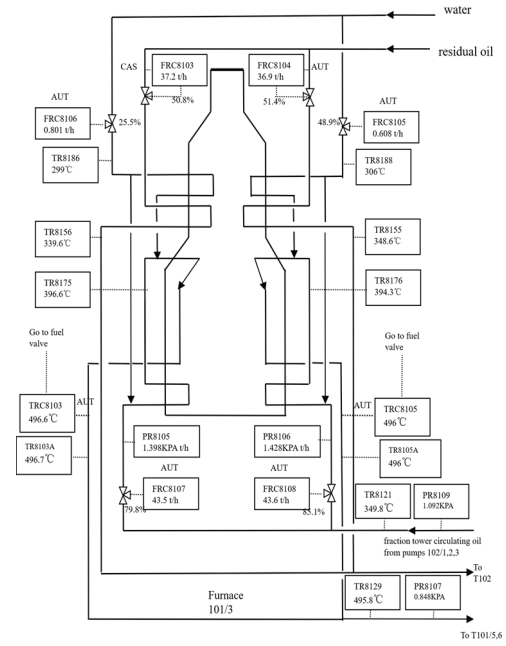


Fig. 4. Industrial coking furnace F101/3 structure.

TABLE II  
SAMPLE SIZE FOR TRAINING AND TESTING

| Data set | Long tail |       | Imbalance |       | Balance |       |
|----------|-----------|-------|-----------|-------|---------|-------|
|          | Normal    | Fault | Normal    | Fault | Normal  | Fault |
| Train    | 500       | 50    | 500       | 250   | 500     | 500   |
| Test     | 100       | 100   | 100       | 100   | 100     | 100   |

The dataset consists of six categories including five typical faults and one normal operation data. The five faults are generated by introducing drift signals, step signals and random signals. Detailed descriptions of the fault types are presented in Table I.

#### B. Experimental Settings and Evaluation Indicators

The sample size used for training and testing is shown in Table II.

The deep learning network models were implemented using the Keras library under the TensorFlow framework in Python. All experiments were conducted on a system equipped with an Intel<sup>1</sup> Core<sup>2</sup> i5-9300H CPU @ 2.40 GHz and standard memory configuration. The proposed DMSAM-OAdB model is compared with six existing methods: focal loss [23], label distribution aware margin (LDAM) loss [24], GC-ResCNN [25], AkNets [26], random forest (RF) algorithm [27], and AdaBoost decision trees (ADTs) [28]. Specifically, the focal loss assigns higher weights to difficult-to-categorize samples; the core idea of LDAM is to adjust the categorization decision boundary based on the number of category samples, giving the minority class a larger decision margin; GC-ResCNN is a

<sup>1</sup>Registered trademark.

<sup>2</sup>Trademarked.

TABLE III  
IMPACT OF THE NUMBER OF BASE CLASSIFIERS ON THE PERFORMANCE OF THE PROPOSED MODEL

| Number  | 1      | 2      | 3      | 4      | 5      | 6      | 7      | 8      |
|---------|--------|--------|--------|--------|--------|--------|--------|--------|
| Acc (%) | 90.50  | 90.67  | 91.17  | 91.67  | 91.50  | 91.33  | 91.17  | 90.83  |
| G-mean  | 0.8966 | 0.8981 | 0.9042 | 0.9101 | 0.9070 | 0.9054 | 0.9036 | 0.8995 |
| Time(s) | 37.37  | 68.16  | 99.78  | 134.40 | 169.30 | 204.18 | 261.88 | 274.80 |

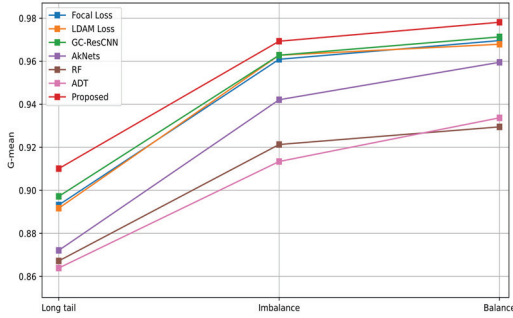


Fig. 5. G-mean results of each model.

global context residual CNN based on a hierarchical architecture, which combines a global context information extraction module with multifeature fusion technology; AkNets generates feature representations with receptive fields of different sizes through multiscale feature fusion and information-guided soft attention mechanism and adaptively adjusts the receptive field size in combination with a dynamic selection strategy; RF is a bagging method based on decision trees and each tree is built on a different training subset with replacement sampling; ADT is an AdaBoost ensemble method that uses decision trees as base classifiers. For the first two methods, we still use the network architecture and training mechanism of the DMSAM-OAdB and only replace the loss function.

In classification tasks with balanced class distributions, accuracy (Acc), precision ( $P$ ), and recall ( $R$ ) are typically adopted as evaluation metrics. When dealing with imbalanced classification, the  $F1$ -score and the  $G$ -mean are more representative. The  $F1$ -score considers the tradeoff between precision and recall and provides a more reasonable reflection of the model's ability to identify minority classes. The  $G$ -mean evaluates the overall performance of the fault diagnosis model by integrating its recognition capability across different fault categories, which emphasizes balanced performance among all classes.

### C. Structural Strategy Analysis

The number of base classifiers significantly affects the performance of the ensemble model. As shown in Table III, under long tail conditions, the performance of DMSAM-OAdB reaches a peak with four base classifiers, showing the highest diagnostic accuracy and  $G$ -mean value. However, increasing the number of base classifiers does not continuously improve the overall model performance as an excessive number of base classifiers may lead to overfitting. Notably,

the OAdB will randomly reset the weights of a certain number of correctly classified majority class samples to zero in the new round of iterations, which will reduce the number of samples participating in training in subsequent iterations and further increases the risk of overfitting of subsequent classifiers. Moreover, the diminishing marginal returns also explain why increasing the number of base classifiers cannot significantly enhance the ensemble model performance, as later base classifiers carry less weight in the overall model and have limited influence on the final decision. Furthermore, the increase in base classifiers leads to higher computational costs and extended training time. Therefore, to achieve a balance between model performance and resource consumption while avoiding overfitting and underfitting, this article selects four DMSAM base classifiers.

### D. Comparison of Existing Methods

Table IV and Fig. 5 demonstrate the results of the proposed DMSAM-OAdB and six comparison methods under three unbalanced conditions. It is observed that the proposed model outperforms the other six comparative models under long tail, imbalanced, and balanced conditions. First, by comparing the performance with the focal loss model and LDAM loss model, we can observe that the DMSAM-OAdB model performs better in terms of both global classification and category balance. Second, in comparison with multiscale attention models such as GC-ResCNN and AkNets, although the former enhances the modeling of local and global discriminative features through hierarchical structure and multifeature fusion mechanisms and the latter integrates multiscale information dynamically via AK convolution and soft squeeze-and-excitation (Soft SE) attention mechanisms, their single-structure nature still limits their effectiveness in handling class imbalance and complex pattern recognition. In contrast, the proposed method leverages the MSAM module to extract critical fault information and incorporates ensemble learning to achieve collaborative training and decision fusion across multiple base classifiers, thereby capturing diverse and fine-grained fault feature distributions more effectively. Third, compared with traditional ensemble learning models such as RF and ADT, the DMSAM-OAdB model enhances fault feature modeling through its integrated feature learning and ensemble strategy and significantly improves diagnostic consistency and reliability under various data distributions. The  $G$ -mean index comprehensively considers the model's classification ability for minority and majority classes. The proposed model reaches the highest  $G$ -mean value of 0.97781 under balanced data distribution and can also reach 0.9101 under extremely imbalanced conditions.

TABLE IV  
FAULT DIAGNOSIS RESULTS IN THREE CONDITIONS

| Model      | Long tail (%) |          | Imbalance (%) |          | Balance (%) |          |
|------------|---------------|----------|---------------|----------|-------------|----------|
|            | Micro F1      | Macro F1 | Micro F1      | Macro F1 | Micro F1    | Macro F1 |
| Focal Loss | 90.33         | 90.43    | 96.17         | 96.15    | 97.00       | 97.00    |
| LDAM Loss  | 90.00         | 90.18    | 96.33         | 96.34    | 96.83       | 96.83    |
| GC-ResCNN  | 90.50         | 90.57    | 96.34         | 93.33    | 97.17       | 97.17    |
| AkNets     | 88.33         | 88.43    | 94.33         | 94.31    | 96.00       | 96.00    |
| RF         | 88.00         | 87.98    | 92.50         | 92.47    | 93.17       | 93.10    |
| ADT        | 87.50         | 88.00    | 91.67         | 91.87    | 93.50       | 93.58    |
| Proposed   | 91.67         | 91.75    | 97.00         | 96.99    | 97.83       | 97.67    |

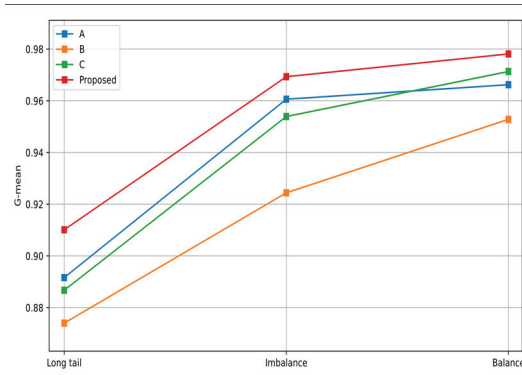


Fig. 6.  $G$ -mean metric under long tail conditions.

TABLE V  
ABLATION EXPERIMENT MODEL SETTINGS

| 模型       | CE | IFL | CNN | MDAM | OAdB |
|----------|----|-----|-----|------|------|
| A        | ✓  |     |     | ✓    | ✓    |
| B        |    | ✓   | ✓   |      | ✓    |
| C        |    | ✓   |     | ✓    |      |
| Proposed |    | ✓   |     | ✓    | ✓    |

### E. Ablation Experiment

In this section, three different models are selected to perform a comparison with the DMSAM-OAdB. The specific settings are shown in Table V, where CNN denotes the  $1 \times 1$  convolutional layer. Among them, Model A modifies the loss function of the proposed model to the CE loss function, Model B replaces the MSAM module with a  $1 \times 1$  convolutional layer, and Model C deletes the OAdB ensemble learning module and uses a single-base learner to make classification predictions. Fig. 6 illustrates the  $G$ -mean metric results of the four models. Fig. 7 presents a heatmap-based visualization of the precision and recall for each fault type under the three imbalance conditions. It can be observed that the proposed models achieve better diagnostic results in all three distribution cases, and with the increase in the number of fault samples, the fault classification ability of the four models has improved. This is because the imbalance of data will complicate the learning of accurate decision boundaries, thereby affecting the

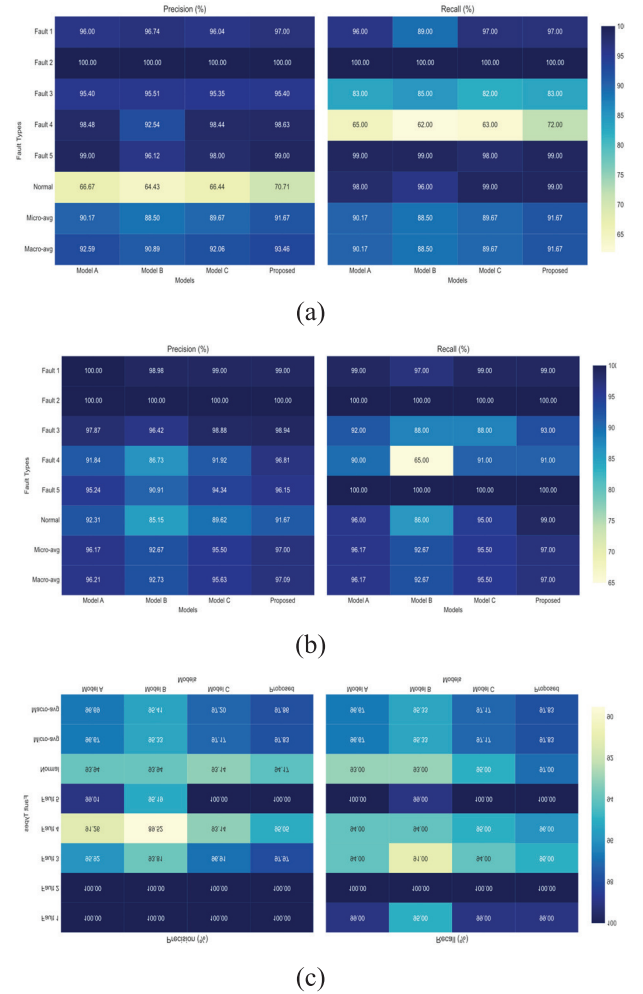


Fig. 7. Classification results under three conditions. (a) Long tail condition. (b) Imbalanced condition. (c) Balanced condition.

classification effect. However, the proposed method integrates the MSAM with the OAdB and introduces the improved loss function IFL to achieve the best diagnostic results under three conditions with finite samples.

## IV. CONCLUSION

This article proposes an end-to-end fault diagnosis framework DMSAM-OAdB for finite and imbalanced samples of

industrial process. The main idea is to utilize MSAMs to extract key features under finite sample and combine improved loss functions and ensemble algorithms to adaptively adjust the attention to minority class samples and difficult-to-classify samples to improve the fault diagnosis ability. The fault diagnosis performance of DMSAM-OAdB was validated using the industrial coking furnace datasets under long tail, imbalanced, and balanced conditions. Results demonstrated that compared with the existing methods, the proposed model has significantly improved the accuracy of fault diagnosis.

## REFERENCES

- [1] N. Wang, F. Yang, R. Zhang, and F. Gao, "Intelligent fault diagnosis for chemical processes using deep learning multimodel fusion," *IEEE Trans. Cybern.*, vol. 52, no. 7, pp. 7121–7135, Jul. 2022.
- [2] Z. Ye and J. Yu, "Multi-level features fusion network-based feature learning for machinery fault diagnosis," *Appl. Soft Comput.*, vol. 122, Jun. 2022, Art. no. 108900.
- [3] K. Liu, N. Lu, F. Wu, R. Zhang, and F. Gao, "Model fusion and multiscale feature learning for fault diagnosis of industrial processes," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6465–6478, Oct. 2023.
- [4] S. Wan, T. Li, B. Fang, K. Yan, J. Hong, and X. Li, "Bearing fault diagnosis based on multisensor information coupling and attentional feature fusion," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–12, 2023.
- [5] Z. Wei, X. Ji, L. Zhou, Y. Dang, and Y. Dai, "A novel deep learning model based on target transformer for fault diagnosis of chemical process," *Process Saf. Environ. Protection*, vol. 167, pp. 480–492, Nov. 2022.
- [6] Q. Wei, X. Tian, L. Cui, F. Zheng, and L. Liu, "WSAFormer-DFFN: A model for rotating machinery fault diagnosis using 1D window-based multi-head self-attention and deep feature fusion network," *Eng. Appl. Artif. Intell.*, vol. 124, Sep. 2023, Art. no. 106633.
- [7] J. Jiang et al., "A digital twin auxiliary approach based on adaptive sparse attention network for diesel engine fault diagnosis," *Sci. Rep.*, vol. 12, no. 1, Jan. 2022, Art. no. 1.
- [8] Q. Zhang et al., "Improved sparse low-rank model via periodic overlapping group shrinkage and truncated nuclear norm for rolling bearing fault diagnosis," *Meas. Sci. Technol.*, vol. 34, no. 6, Jun. 2023, Art. no. 065009.
- [9] Q. Zhu, X. Wang, N. Zhang, Y. Xu, and Y. He, "Novel K-medoids based SMOTE integrated with locality preserving projections for fault diagnosis," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–8, 2022.
- [10] M. Qian and Y.-F. Li, "A novel adaptive undersampling framework for class-imbalance fault detection," *IEEE Trans. Rel.*, vol. 72, no. 3, pp. 1003–1017, Sep. 2023.
- [11] P. Peng, W. Zhang, Y. Zhang, Y. Xu, H. Wang, and H. Zhang, "Cost sensitive active learning using bidirectional gated recurrent neural networks for imbalanced fault diagnosis," *Neurocomputing*, vol. 407, pp. 232–245, Sep. 2020.
- [12] Z. Mian et al., "A literature review of fault diagnosis based on ensemble learning," *Eng. Appl. Artif. Intell.*, vol. 127, Jan. 2024, Art. no. 107357.
- [13] F. Duan, S. Zhang, Y. Yan, and Z. Cai, "An oversampling method of unbalanced data for mechanical fault diagnosis based on MeanRadius-SMOTE," *Sensors*, vol. 22, no. 14, p. 5166, Jul. 2022.
- [14] J. Zhou, Z. Wu, Q. Wang, and Z. Yu, "Fault diagnosis method of smart meters based on DBN-CapsNet," *Electronics*, vol. 11, no. 10, p. 1603, May 2022.
- [15] X. Yuan et al., "Quality prediction modeling for industrial processes using multiscale attention-based convolutional neural network," *IEEE Trans. Cybern.*, vol. 54, no. 5, pp. 2696–2707, May 2024.
- [16] G. Wang, Y. Li, Y. Wang, Z. Wu, and M. Lu, "Bidirectional shrinkage gated recurrent unit network with multiscale attention mechanism for multisensor fault diagnosis," *IEEE Sensors J.*, vol. 23, no. 20, pp. 25518–25533, Oct. 2023.
- [17] M. Zhao, S. Zhong, X. Fu, B. Tang, and M. Pecht, "Deep residual shrinkage networks for fault diagnosis," *IEEE Trans. Ind. Informat.*, vol. 16, no. 7, pp. 4681–4690, Jul. 2020.
- [18] M. A. Ganaie, M. Hu, A. K. Malik, M. Tanveer, and P. N. Suganthan, "Ensemble deep learning: A review," *Eng. Appl. Artif. Intell.*, vol. 115, Sep. 2022, Art. no. 105151.
- [19] W. Tao, Z. Sun, Z. Yang, B. Liang, G. Wang, and S. Xiao, "Transformer fault diagnosis technology based on AdaBoost enhanced transferred convolutional neural network," *Expert Syst. Appl.*, vol. 264, Mar. 2025, Art. no. 125972.
- [20] B. S. Raghuwanshi and S. Shukla, "Classifying imbalanced data using BalanceCascade-based kernelized extreme learning machine," *Pattern Anal. Appl.*, vol. 23, no. 3, pp. 1157–1182, Aug. 2020.
- [21] Y. Xi, M. Li, F. Zhou, X. Tang, Z. Li, and J. Tian, "SE-inception-ResNet model with focal loss for transmission line fault classification under class imbalance," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–17, 2024.
- [22] R. Zhang, A. Xue, and F. Gao, "Temperature control of industrial coke furnace using novel state space model predictive control," *IEEE Trans. Ind. Informat.*, vol. 10, no. 4, pp. 2084–2092, Nov. 2014.
- [23] T.-Y. Lin, P. Goyal, R. B. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.
- [24] K. Cao, C. Wei, A. Gaidon, N. Arechiga, and T. Ma, "Learning imbalanced datasets with label-distribution-aware margin loss," in *Proc. 33rd Int. Conf. Neural Inf. Process. Syst.*, 2019, pp. 1567–1578.
- [25] Y. Xu, X. Yan, B. Sun, and Z. Liu, "Global contextual residual convolutional neural networks for motor fault diagnosis under variable-speed conditions," *Rel. Eng. Syst. Saf.*, vol. 225, Sep. 2022, Art. no. 108618.
- [26] Y. Chang, J. Chen, Q. Chen, S. Liu, and Z. Zhou, "CFs-focused intelligent diagnosis scheme via alternative kernels networks with soft squeeze-and-excitation attention for fast-precise fault detection under slow & sharp speed variations," *Knowl.-Based Syst.*, vol. 239, Mar. 2022, Art. no. 108026.
- [27] N. Haque, A. Jamshed, K. Chatterjee, and S. Chatterjee, "Accurate sensing of power transformer faults from dissolved gas data using random forest classifier aided by data clustering method," *IEEE Sensors J.*, vol. 22, no. 6, pp. 5902–5910, Mar. 2022.
- [28] Y. Xu et al., "A multi-fault diagnosis method based on improved SMOTE for class-imbalanced data," *Can. J. Chem. Eng.*, vol. 101, no. 4, pp. 1986–2001, Apr. 2023.